

SMART INDIA HACKATHON 2026



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HACKATHON
2026

LabelJaano

Regulatory compliance for packaged-food labels, from a photograph

- Problem Statement ID – _____
- Problem Statement Title- _____
- Theme- _____
- PS Category- Software
- Team ID- _____
- Team Name (Registered on portal)- _____

BUILT AND TESTED

8 versioned rulepacks · 48 declarations · 79 machine-checked clauses

12 lab-only limits cited but never scored · 308 tests passing



IDEA TITLE

Proposed Solution (Describe your Idea/Solution/Prototype)

LabelJaano turns a photograph of a packaged-food label into a clause-cited compliance verdict — the regulation itself lives as versioned data, not as code.

Detailed explanation of the proposed solution

- Photograph the label panels; a reference card or ArUco marker in frame fixes real-world scale
- A vision-LLM extracts only the declarations the loaded rulepacks ask for — the prompt is generated from those packs
- PaddleOCR glyph boxes plus the marker yield a true millimetre font height, measured independently of the LLM
- A declarative rule engine scores each declaration and cites the clause behind every finding

How it addresses the problem

- Enforcement today is manual and sample-based — one inspector, one packet at a time
- Violations are small by nature: a missing veg mark, a net-quantity font a millimetre under the mandated height
- Puts a gazette rule in the hands of a shopper in the aisle and an officer in the field, in seconds
- Officer scan history and CSV export make findings reviewable rather than one-off

Innovation and uniqueness of the solution

- 8 versioned JSON rulepacks → 48 declarations → 79 checks, with zero regulation-specific branches in the scoring path
- A category pack overrides a base rule by reusing its id: scan as wine and the MRP “inclusive of all taxes” check disappears, per the Rule 6(1)(e) proviso
- Two-tier design — lab-only limits are cited, never scored, so they structurally cannot move a verdict
- Add a pack, POST /reload, new rules enforced with no code change

8 rulepacks

FSSAI + Legal Metrology, each versioned JSON

79 checks

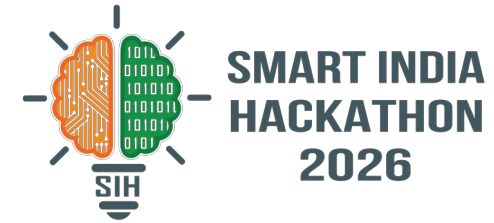
48 declarations, 6 check types, 7 validators

308 tests

full suite passing, CI on Python 3.11 and 3.12

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TECHNICAL APPROACH



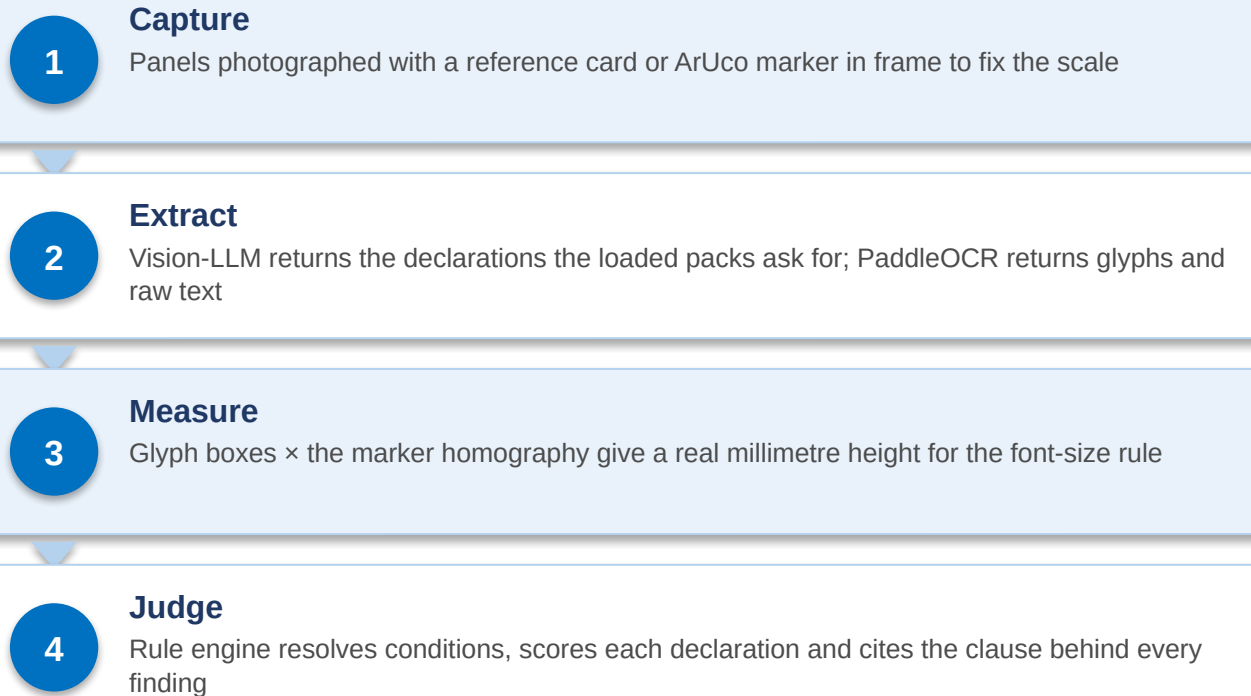
Technologies to be used (e.g. programming languages, frameworks, hardware)

- **Backend** — Python 3.11 / 3.12, FastAPI, SQLite in WAL mode
- **Auth** — stdlib only — PBKDF2-HMAC-SHA256 at 600,000 iterations, purpose-scoped HS256 tokens
- **Vision** — Gemini vision API for field extraction from the label panels
- **OCR** — PaddleOCR for glyph bounding boxes and raw text
- **Calibration** — OpenCV ArUco / reference-card homography for a pixel-to-millimetre scale
- **Rules** — versioned JSON rulepacks — 6 check types, 7 validators, a 20-token condition vocabulary (18 used by the packs today)
- **Mobile** — Flutter client for capture and verdict display
- **Ops** — Docker as a non-root user, GitHub Actions CI, 308 tests

Prototype status

- FastAPI service and Flutter client run end to end today
- Ships as one non-root Docker image over a SQLite volume — no GPU required
- Rulepacks hot-reload over POST /reload without a restart

Methodology and process for implementation (Flow Charts/Images/ working prototype)



Honest about the evidence: field values come from the LLM read; font height is the one check measured independently, and it returns SKIP — never a guess — with no marker in frame.

Analysis of the feasibility of the idea — we separate what a camera can prove from what only a laboratory can settle

Scored from the photograph — feasible today

- Net quantity, unit and the MRP “inclusive of all taxes” wording
- FSSAI licence number and logo; veg / non-veg mark
- Date marking — manufacture, best before, use by
- Ingredients, additives, allergen declarations
- Font height in mm against Rule 7(2) Table-I

Cited, never scored — a laboratory decides

- Heavy-metal and contaminant ceilings
- Aflatoxin and mycotoxin limits
- Alcohol-by-volume tolerance
- Fortification micronutrient ranges
- 12 such limits ride a separate data path with 0 scored checks

Potential challenges and risks

- Extraction is a single vision-LLM call — a hallucinated field becomes a finding
- Veg mark and FSSAI logo are critical checks but rest on LLM self-report, with no computer-vision detector yet
- Some format checks are permissive regexes a lookalike string can satisfy
- No marker in frame means no font-height verdict
- Rulepacks go stale as the gazette is amended

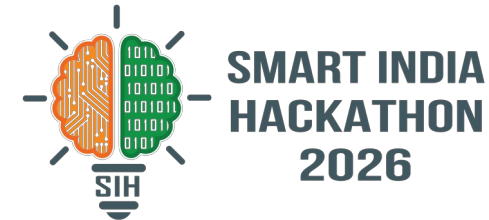
Strategies for overcoming these challenges

- Cross-check the OCR text against the LLM read and surface disagreement instead of merging the two
- Drop a CV detector in behind the existing check-type registry — the seam is already there, no engine change
- Tighten regexes with word boundaries and keyword context; add an adversarial negative-match test suite
- Return SKIP, never a score, wherever evidence is missing — already the rule for font height
- Packs are versioned JSON, so a gazette diff is a data review, not a release

Viability — one container plus a SQLite file; no GPU and no per-seat licence. Anonymous scanning needs no account, officer features sit behind purpose-scoped tokens, and the suite runs in CI on Python 3.11 and 3.12.

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IMPACT AND BENEFITS



Potential impact on the target audience

C

Consumers

Apply a rule they cannot read — allergens, veg status, net quantity, MRP — while standing in the aisle, in seconds.

O

Legal Metrology & FSSAI officers

Screen many packets in the time one manual inspection takes; every scan is stored with its clause citations and exports to CSV.

B

Brands and packaging teams

Catch a font-height or declaration defect at artwork stage, before a print run and before a seizure.

Benefits of the solution (social, economic, environmental, etc.)

- **Social** — Allergen and veg / non-veg declarations become verifiable rather than trusted — the checks that matter most to the people least able to audit them.
- **Economic** — A recall, a seizure or a reprint costs far more than a pre-print check. Compliance moves from after the fact to before the press run.
- **Environmental** — Fewer scrapped packaging runs and fewer reprints, because the defect is caught in artwork rather than on the shelf.
- **Governance** — Every finding names the clause it came from, so a verdict can be contested on the record instead of argued from opinion.
- **Scale** — Enforcement coverage stops being limited by the number of inspectors available to read packets by eye.
- **Deterrence** — A violation anyone can find in seconds is a violation worth not committing — the expectation shifts before the first penalty is issued.

79 checks per scan

every one carrying the clause it came from

12 lab-only limits

cited for context, structurally never scored

0 accounts needed

anonymous scanning; officer tools behind tokens

Details / Links of the reference and research work

Statutory sources — each encoded as a versioned rulepack

- **Legal Metrology (Packaged Commodities) Rules, 2011** — Rule 6 declarations, the Rule 6(1)(e) alcohol proviso and the Rule 7(2) Table-I font heights, as amended by G.S.R. 629(E) — legalmetrology.gov.in
- **FSS (Labelling and Display) Regulations, 2020** — mandatory particulars and display requirements — fssai.gov.in
- **FSS (Packaging and Labelling) Regulations, 2011** — product-specific declarations gated on traits such as added colour or caffeine
- **FSS (Contaminants, Toxins and Residues) Regulations, 2011** — the reference-only tier: 5 limits, 0 scored declarations
- **FSS (Alcoholic Beverages Standards) Regulations, 2018** — Part-5 labelling provisions, including the ABV tolerance
- **FSS (Fortification of Foods) Regulations, 2018** · **FSS (Organic Foods) Regulations, 2017** · **FSS (Health Supplements, Nutraceuticals ...) Regulations, 2016** — trait-gated packs, applied only when the label makes the claim
- **FSSAI Food Safety Compliance System (FoSCoS)** — the 14-digit licence number format — foscoss.fssai.gov.in

Technical and comparative references

- **PaddleOCR** — text detection and glyph bounding boxes — github.com/PaddlePaddle/PaddleOCR
- **OpenCV ArUco** — fiducial marker detection and homography for metric calibration — docs.opencv.org
- **Google Gemini vision API** — structured field extraction from label photographs — ai.google.dev
- **FastAPI** — the service layer — fastapi.tiangolo.com
- **Codex Alimentarius STAN 1-1985** — General Standard for the Labelling of Prepackaged Foods
- **EU Regulation 1169/2011, Article 13** — the 1.2 mm x-height minimum — the international analogue of Rule 7(2)
- **NIST SP 800-63B** · **RFC 8018** — memorised-secret storage and PBKDF2 — the basis for 600,000 iterations
- **RFC 7519** · **RFC 8725** — JWT and JWT best current practice — why the algorithm is pinned and expiry is mandatory

Note on the packs — each pack carries a source_note recording which values are verbatim from the Rules and which regexes are engineered starting points. The packs are an enforcement aid, not a legal opinion.